

Stochastic Modeling and Simulation
3rd Y - Semester 5
2025-2026

Test - 50 minutes

Exercise 1 *Simulation of the game “Heads” or “Tails” played by player A, who wins 10 DA if the coin lands on ‘Heads’ and loses 10 DA if it lands on “Tails.” The simulation will involve 10 tosses, assuming that $p = \mathbb{P}(\text{Heads}) = \frac{3}{4}$ and $\mathbb{P}(\text{Tails}) = \frac{1}{4}$*

1. *How much does the player win?*
2. *Now suppose that the coin is balanced (not rigged). How much will the player win after 10 tosses?*

Exercise 2 *Consider the triangular random variable X with pdf*

$$f(x) = \begin{cases} 0 & \text{if } x < 2a \text{ or } x \geq 2b \\ \frac{x-2a}{(b-a)^2} & \text{if } 2a \leq x < a+b \\ \frac{2b-x}{(b-a)^2} & \text{if } a+b \leq x < 2b \end{cases}$$

1. *Derive the corresponding cdf F .*
2. *Using the inverse transform method, present a simulation algorithm for r.v X . Then, for $a = 1$ and $b = 2$, generate three observations of this variable.*

For applications, use the following sequence of random numbers:

0.404190 0.735378 0.369251 0.705100 0.833575
0.397475 0.151053 0.925697 0.928252 0.481475